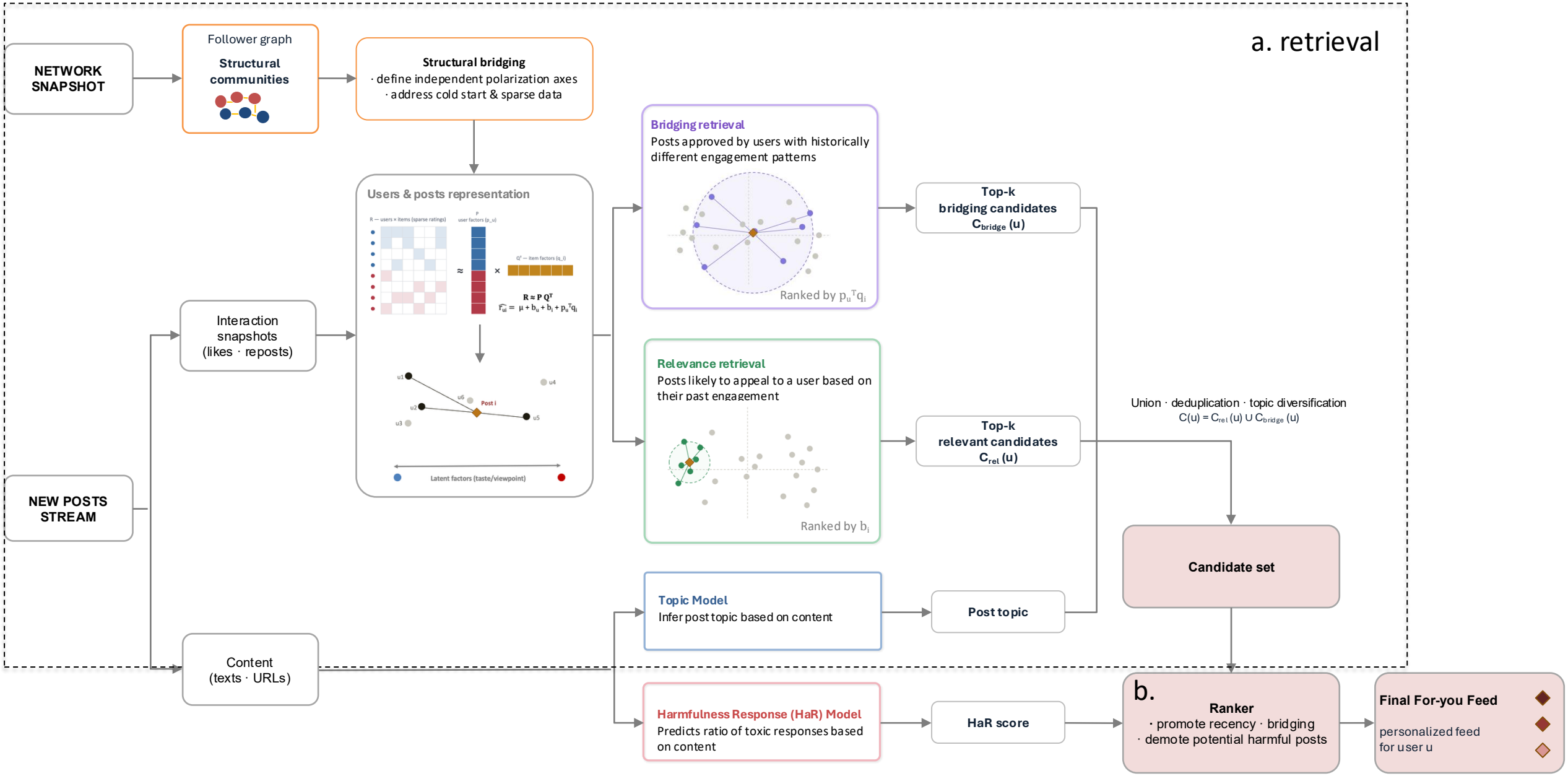

a. retrieval
NETWORK SNAPSHOT
Follower graph
Structural communities
Structural bridging
· define independent polarization axes
· address cold start & sparse data
Users & posts representation
R — users × items (sparse ratings)
P user factors (p_u)
Q^T — item factors (q_i)
R ≈ P Q^T
r̂_ui = μ + b_u + b_i + p_u^T q_i
u1
u2
u3
u4
u5
u6
Post i
Latent factors (taste/viewpoint)
Interaction snapshots (likes · reposts)
NEW POSTS STREAM
Content (texts · URLs)
Bridging retrieval
Posts approved by users with historically different engagement patterns
Ranked by p_u^T q_i
Top-k bridging candidates C_bridge (u)
Relevance retrieval
Posts likely to appeal to a user based on their past engagement
Ranked by b_i
Top-k relevant candidates C_rel (u)
Union · deduplication · topic diversification
C(u) = C_rel (u) ∪ C_bridge (u)
Candidate set
Topic Model
Infer post topic based on content
Post topic
Harmfulness Response (HaR) Model
Predicts ratio of toxic responses based on content
HaR score
b.
Ranker
· promote recency · bridging
· demote potential harmful posts
Final For-you Feed
personalized feed for user u